TIMEZONE ESP 32 MATRIX CLOCK

MINI PROJECT REPORT

24MNC 120 DESIGN AND ENGINEERING

Submitted

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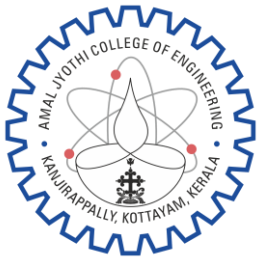
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IN

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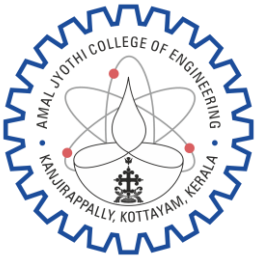


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BONA FIDE CERTIFICATE

This is to certify that the Mini Project Report entitled "TIMEZONE ESP 32 MATRIX CLOCK" **submitted by LEVIN BINOY, (17634), MEEVAL ANNA MANOJ (16823), JOSWIN JOSEPH (17629), JOEL JACOB K (17625) , in partial fulfillment of the requirement for the award of the Bachelor of Technology in ELECTRONICS AND COMMUNICATION is a bona fide record of the work carried out at AMAL JYOTHI COLLEGE OF ENGINEERING (AUTONOMOUS), under our guidance and supervision.**

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ACKNOWLEDGMENT

We express our sincere gratitude to the Almighty for blessing us with the knowledge, strength, and opportunity to successfully complete this mini project. We sincerely thank our project guide for the invaluable guidance, continuous support, and constructive suggestions that helped us at every stage of this project. We also thank all the faculty members of the Department of Electronics and Communications for their timely advice and inspiration. We thank our parents for having sacrificed a lot in their lives to impart the best education to us and for making us promising professionals of tomorrow. We extend our sincere gratitude to our beloved friends and well-wishers for their invaluable prayers and powerful motivation throughout this project.

ABSTRACT

This project details the design and implementation of a reliable, maintenance-free digital clock that utilizes Internet of Things (IoT) connectivity to automatically synchronize and maintain highly accurate time. Traditional digital clocks often suffer from manual time drift and require tedious reconfigurations after power outages. To eliminate these issues, this system leverages Network Time Protocol (NTP) over Wi-Fi to fetch real-time data, utilizing background resynchronization routines to maintain precise atomic time keeping without disrupting the device's display functionality.

A core focus of the design is user accessibility and operational robustness. Rather than hardcoding network credentials into the firmware, the system features an on-device Wi-Fi manager configuration mode. Upon initial boot or network loss, the device hosts a local captive portal web server, allowing users to securely input local Wi-Fi credentials via any smartphone or computer. These settings are then saved to persistent, non-volatile onboard storage Flash, ensuring the clock seamlessly reconnects and resumes operation after a power cycle. The resulting prototype offers a standalone, consumer-ready appliance that merges the accuracy of network-synced time with a self-healing, user-configurable architecture.

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Chapter 1

Introduction

1.1 Scope of the Project

In the modern Internet of Things (IoT) era, accurate timekeeping remains a fundamental requirement for smart environments, industrial automation, and everyday consumers. Traditional digital clocks rely heavily on internal Real-Time Clock (RTC) modules. While RTCs are functional, they suffer from natural crystal oscillator drift over time, requiring manual adjustments to maintain precision. Furthermore, tasks such as handling daylight-saving time transitions or manual updating after prolonged power outages make legacy digital clocks high-maintenance devices.

1.2 Background and Motivation

The motivation behind the ESP32 IoT Matrix Clock is to eliminate manual intervention entirely by leveraging ubiquitous Wi-Fi connectivity. By utilizing the Network Time Protocol (NTP), the clock syncs directly with atomic-accuracy time servers across the internet, ensuring down-to-the-second precision without the need for an expensive dedicated hardware RTC. Designed around the cost-effective and highly powerful ESP32 microcontroller, this project merges reliable network-synchronized timekeeping with high-visibility, customizable industrial LED displays.

1.3 Problem statement

Existing DIY and consumer-grade digital clocks present several core operational challenges:

- **Hardcoded Credentials: Many open-source IoT clocks require users to hardcode their Wi-Fi SSID and password directly into the firmware,**

meaning the clock becomes non-functional if moved to a new network unless re-flashed via a computer.

- **Electrical Noise and Static: Tactile buttons in DIY micro-controller environments are highly susceptible to environmental static and electromagnetic interference (EMI), leading to "ghost presses" that erratically change the clock's mode.**
- **Poor Distance Visibility: Standard 7-segment displays or stock thin digital fonts often struggle with legibility from a distance or at sharp angles.**
- **Blocking Network Calls: Poorly optimized firmware will freeze the entire display and time-tracking system if the local Wi-Fi router drops its connection, rendering the clock useless during network updates.**

Chapter 2

Literature review

2.1 Design Objectives

The primary objectives of this project are:

1. **Automated Maintenance-Free Operation: Achieve automated time fetching and background resynchronization without blocking user interaction or freezing the display.**
2. **Dynamic Provisioning: Provide a secure, browser-based web interface (Captive Portal) directly hosted on the ESP32 to modify, add, or delete Wi-Fi networks dynamically.**
3. **Enhanced Visual Presentation: Create a high-visibility interface using a custom bold font tailored specifically for a multi-module MAX7219 matrix.**
4. **Hardware Rigidity: Enforce input line stability to neutralize false triggers from static or inductive loads nearby.**

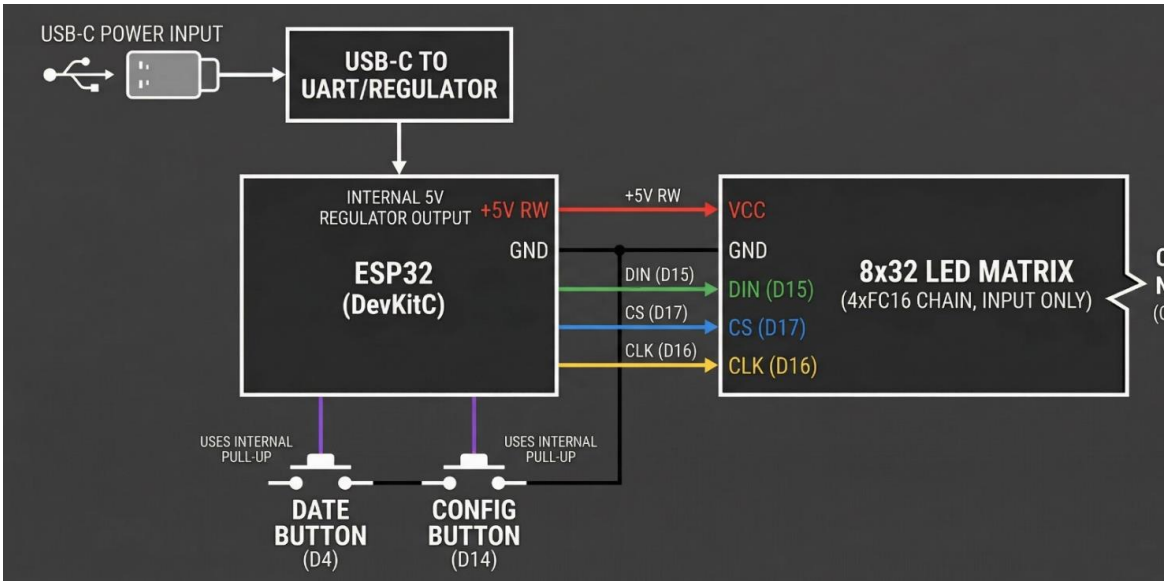
2.2 Design Constraints

- **Microcontroller Constraints: The software must operate efficiently within the memory boundaries of the ESP32, specifically caching custom typography structures to flash memory (PROGMEM) to save runtime SRAM.**
- **Network Independence: Local timekeeping and UI rendering must operate completely unhindered, relying on a non-blocking asynchronous state machine when network drops occur.**
- **Power & Wiring Limits: The hardware layout must ensure a common ground loop to eliminate noise across the chained MAX7219 matrices, drawing power from a stable 5V rail capable of sustaining high-current LED bursts**

Chapter 3

Prototype Details (Including Drawings and Prototype Images with description)

3.1 Schematic & Hardware Connectivity



[Figure 3.1: Block Diagram of the System/ Circuit Diagram*]*

3.2 Components Used

Component	Function	Defined Constant	ESP32 GPIO	Notes
Status LED	Built-in RGB	INTERNAL_LED_PIN	**GPIO 8**	NeoPixel/WS2812 type
MAX7219 Matrix	SPI Data	DATA_PIN	**GPIO 15**	
	SPI Clock	CLK_PIN	**GPIO 20**	
	SPI Chip Select	CS_PIN	**GPIO 7**	
Date Button	Input	DATE_BTN_PIN	**GPIO 4**	Configured as INPUT_PULLUP
Config Button	Input	CONFIG_BTN_PIN	**GPIO 14**	Configured as INPUT_PULLUP

Table 3.1: Components and Their Functions

Operational Modes Breakdown

Normal Clock Mode (Default Operational State)

- **Time & Date Display: Displays time using a custom ultra-bold font on a 4-panel MAX7219 matrix. Pressing the Date Button interrupts the clock to scroll the full day and date for 10 seconds before returning.**
- **Non-Blocking Background Sync: Every 6 hours, the clock updates its time via NTP and auto-detects time zones using an HTTP API. This runs on a background state machine to prevent display stuttering.**
- **Power & Noise Optimization: The Wi-Fi radio turns completely OFF immediately after syncing to minimize power consumption and eliminate electromagnetic noise on the LED display.**
- **LED Status Monitor: An onboard Neo Pixel glows Green during normal operation and shifts to Red during an active background network sync.**

Web Configuration Mode (Setup State)

- **Trigger: Activated by holding the Config Button for 1 second (indicated by a Blue NeoPixel and a "SET WIFI" display message).**
- **Captive Portal: The ESP32 acts as a standalone Wi-Fi Access Point ("ESP32-Matrix-Clock"). It runs a DNS captive portal that automatically redirects any connected device to the local settings page (192.168.4.1).**
- **Web Dashboard: Provides a dark-themed UI to scan/add/delete multiple Wi-Fi networks and adjust display brightness dynamically using a slider (0–15 intensity levels). Settings are saved directly to non-volatile memory (Preferences).**

Chapter 4
Conclusion

The ESP32 IoT Matrix Clock successfully demonstrates an integrated approach to network-synchronized, maintenance-free timekeeping. By pairing the high-performance ESP32 with an optimized background state machine, the system effectively resolves the issues of display-freezing and manual clock adjustment common in older digital clock models.

Key design elements such as the custom-designed ultra-bold typography for structural legibility, an on-chip web portal for seamless credential provisioning, and rigorous software-level input debouncing elevate the device from a basic hobbyist project to a reliable, consumer-ready appliance. The physical hardware assembly adheres strictly to its designated SPI configuration, ensuring clear data distribution across the multi-panel LED matrices without signal degradation. Ultimately, this project proves that leveraging standard IoT frameworks can transform everyday timekeeping into a highly reliable, accurate, and user-friendly experience.

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